

# KNN — Model Evaluation

| field               | value                |
|---------------------|----------------------|
| model_label         | KNN                  |
| estimator_class     | KNeighborsClassifier |
| param.algorithm     | auto                 |
| param.leaf_size     | 30                   |
| param.metric        | minkowski            |
| param.metric_params | None                 |
| param.n_jobs        | None                 |
| param.n_neighbors   | 15                   |
| param.p             | 2                    |
| param.weights       | uniform              |

KNN — SUMMARY OUTPUT (Classification)

| Metric            | Value                     |
|-------------------|---------------------------|
| Test observations | 2360                      |
| Accuracy          | 0.4924                    |
| Balanced accuracy | 0.4463                    |
| Macro precision   | 0.4352                    |
| Macro recall      | 0.4463                    |
| Macro F1          | 0.4275                    |
| Weighted F1       | 0.4646                    |
| Dominant error    | 1 (Draw) → 0 (Loss) [289] |

# KNN — Metrics Summary

| metric             | value               |
|--------------------|---------------------|
| accuracy           | 0.4923728813559322  |
| balanced_accuracy  | 0.4462556589537223  |
| precision_macro    | 0.43523248366859707 |
| recall_macro       | 0.4462556589537223  |
| f1_macro           | 0.4274865649953277  |
| precision_weighted | 0.4586982486952801  |
| recall_weighted    | 0.4923728813559322  |
| f1_weighted        | 0.46464897094767166 |
| n_test             | 2360.0              |

KNN — Classification Report

| class_or_avg | precision           | recall              | f1-score            | support            |
|--------------|---------------------|---------------------|---------------------|--------------------|
| 0            | 0.5048034934497817  | 0.6450892857142857  | 0.5663890249877511  | 896.0              |
| 1            | 0.26639344262295084 | 0.11443661971830986 | 0.16009852216748768 | 568.0              |
| 2            | 0.5345005149330587  | 0.5792410714285714  | 0.5559721478307446  | 896.0              |
| accuracy     | 0.4923728813559322  | 0.4923728813559322  | 0.4923728813559322  | 0.4923728813559322 |
| macro avg    | 0.43523248366859707 | 0.4462556589537223  | 0.4274865649953277  | 2360.0             |
| weighted avg | 0.4586982486952801  | 0.4923728813559322  | 0.46464897094767166 | 2360.0             |

KNN — Confusion Matrix

